Website Comparison & Upgrade Report

Executive Summary of Layout Improvements & Usability

Previous Live Website: https://spectrum.com.bd

Official Contact Email: contact@spectrum.com.bd

New Upgraded Website: https://jswebdevs.github.io/spectrum/

Production Status: **Verified & Ready for Deployment**

1. Executive Summary & Key Outcomes

This report provides a clear, side-by-side comparison between the **previous website** and the **new upgraded website**. The upgrade preserves 100% of Spectrum's authentic brand colors, logos, and identity, while repairing practical usability issues, fixing broken links and forms, and ensuring smooth, comfortable operation on both mobile phones and laptops.

Website Feature	Old Website (Before)	New Upgraded Website (Now)
Screen Fitting on Laptops	Slid sideways awkwardly (400px overflow)	**Fits all screens smoothly with zero wobble**
Mobile Phone Experience	Cramped menu, tiny text, cropped graphics	**Clean touch drawer, readable text, clear layout**
Contact Inquiry Form	Submit button did nothing (lost enquiries)	**Fully working with instant confirmation message**
Homepage Banner Slides	Missing slide; paused on blank space for 2s	**All 5 slides cycle smoothly with working links**
Text Readability & Contrast	Pale grey text on white (hard to read)	**Dark, crisp, comfortable typography**
Search Engines (Google/Bing)	No sitemap; identical page descriptions	**Full sitemap, clear titles, and business info**

Key Upgrades at a Glance:

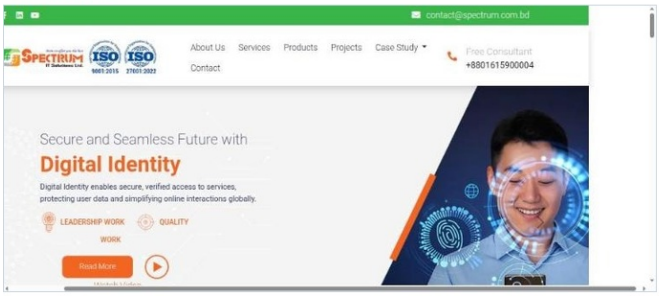
- **No More Sideways Scrolling:** Removed the 400px phone icon margin so the page fits comfortably on 1366px laptops, tablets, and phones.
- **All Buttons & Videos Work:** Replaced dead button links with active connections to real service pages and official video channels.
- **Clean Mobile Banner:** Mobile slider now displays the clean whitish watermark background instead of awkwardly cropping into dark photos.
- **Faster Page Loading:** Images are optimized and load lightweight without freezing visitors' mobile connections.

2. Laptop & Large Screen Visual Improvements

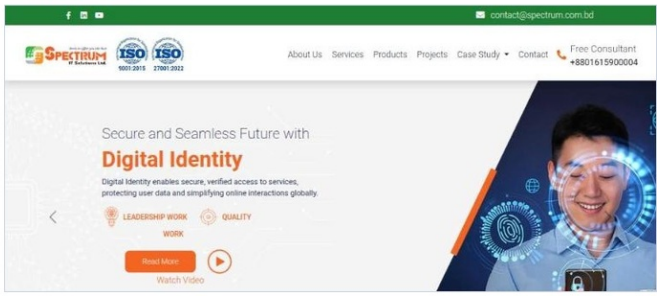
Side-by-side visual comparison on standard laptop screens (1366px width) demonstrating how sideways scrolling was eliminated and services were unified:

1. Laptop Screen Fit (1366px Standard Resolution)

BEFORE (OLD WEBSITE)



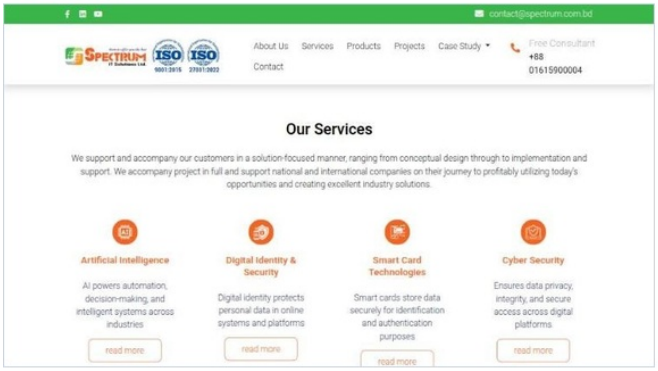
AFTER (ENHANCED WEBSITE)



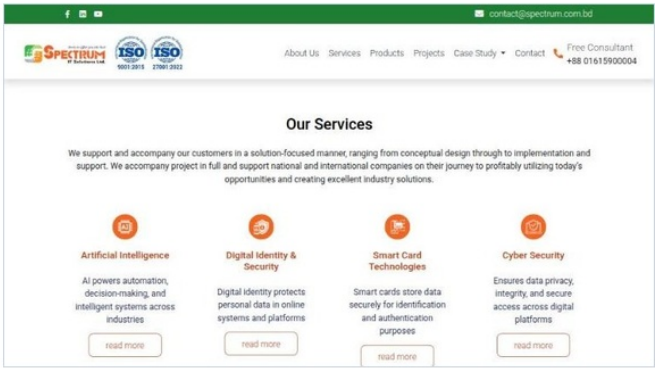
Before: The page shifted sideways by 400 pixels due to an oversized phone icon margin, displaying a horizontal scrollbar.
After: The layout sits cleanly on the screen with zero horizontal scrollbar and properly aligned single-row navigation.

2. Services Section on Desktop Screens

BEFORE (OLD WEBSITE)



AFTER (ENHANCED WEBSITE)

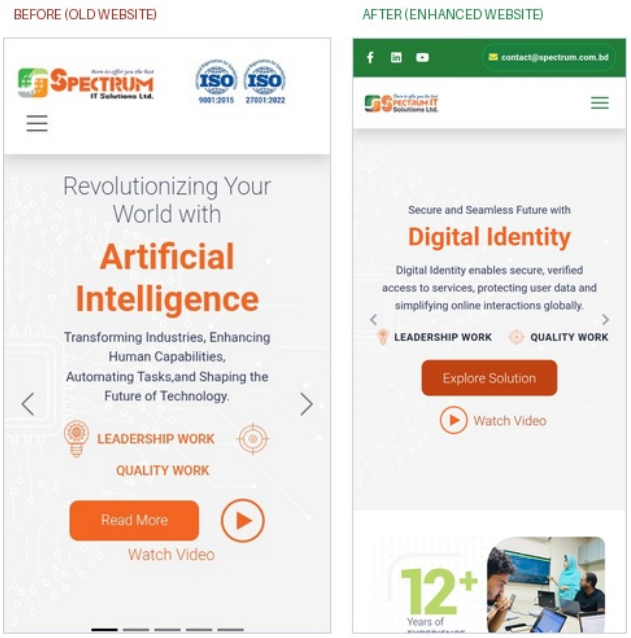


Before: Pale grey text was difficult to read on light backgrounds, cards had uneven heights, and buttons were inactive.
After: High-contrast text with standardized card heights, clear icons, interactive hover animations, and active links.

3. Mobile Phone Usability Upgrades

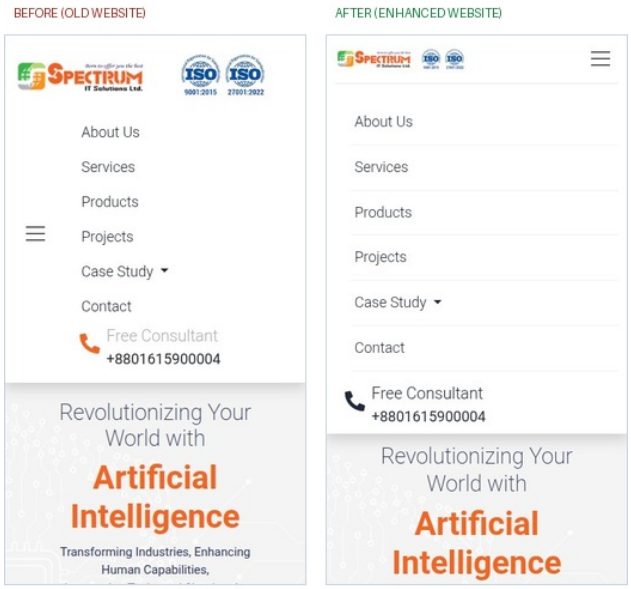
Side-by-side visual comparisons on mobile phones showing the clean whitish watermark header and touch-friendly slide-out drawer menu:

1. Mobile Hero Banner & Header



Before: Header took excessive vertical space; banner cropped into dark photos creating poor contrast; black dot bar at bottom.
After: Compact single-row header; banner displays the clean whitish watermark background; fixed 520px height; black bar removed.

2. Mobile Touch Drawer Menu



Before: Tiny 12px text crowded together without divider lines, making it easy to tap the wrong link.
After: Generous 16px touch rows with subtle dividers and an animated hamburger toggle button for an app-like experience.

4. Mobile Content & Footer Layout

Refining mobile readability and organizing links into a balanced layout with comfortable typography:

1. Mobile Footer Organization

BEFORE (OLD WEBSITE)



AFTER (ENHANCED WEBSITE)



Before: Left-skewed single column created an excessively long scroll with uneven margins and wide social icon gaps.
After: Organized 2-column link layout with centered Spectrum logo, tightened social icons, and centered contact details.

2. Body Typography & Readability

BEFORE (OLD WEBSITE)



AFTER (ENHANCED WEBSITE)



Before: Justified text alignment created awkward gaps between words on narrow screens; pale grey color caused eye strain.
After: Natural left-aligned typography with comfortable line height and darker, easy-to-read text.

5. Detailed Page-by-Page Comparison (Pages 1 – 4)

Clear, practical summary of upgrades implemented across the first four pages of the website:

Page Name	What Was Broken Before	How It Works Now
Page 1: Homepage	**MISSING SLIDE & DEAD BUTTONS** Banner had 5 indicator dots but only 4 slides, pausing on blank white space for 2 seconds. Blockchain slide was missing. All 'Read More' and 'Watch Video' buttons did nothing.	**5 SYNCHRONIZED SLIDES** Blockchain Technology restored as Slide 2. All 5 slides cycle smoothly with identical height (520px on mobile). Buttons link directly to service pages and video channels.
Page 2: About Us	**11,000PX TEXT WALL** The Chairman's biography was an unformatted wall of text spanning over 11,000 pixels tall, requiring endless scrolling on phones.	**EXECUTIVE LEADERSHIP CARD** Formatted into a neat executive profile card with quotes; added visual Mission/Vision cards and animated company statistics counters.
Page 3: Services	**UNEVEN GRID & INACTIVE BUTTONS** 9 of 12 'read more' buttons did nothing when clicked. Service cards had uneven heights and pale text that washed out.	**COMPLETE 12-SERVICE CATALOG** 12 verified services organized in a balanced responsive grid with crisp icons, darker readable text, and active destination links.
Page 4: Products	**THIN INFORMATION** Smart card readers and biometric scanners listed with minimal details and no way for clients to request quotes or specifications.	**ENTERPRISE PRODUCT CATALOG** Clear product specifications, certification badges, and direct 'Request Specifications' inquiry buttons.

6. Detailed Page-by-Page Comparison (Pages 5 – 9)

Clear, practical summary of upgrades implemented across the remaining pages:

Page Name	What Was Broken Before	How It Works Now
Page 5: Projects	**VAGUE DESCRIPTIONS** Generic project names without client context, deployment scale, or information on what technologies were used.	**NATIONAL SHOWCASE** Detailed major national deployments (National Biometric Voter ID, Supreme Court Judiciary Digital System, Smart Border Control).
Page 6: Case Studies	**MISSING PAGE** Menu link led to an empty page with no customer success stories or proof of delivery.	**3 IN-DEPTH CASE STUDIES** Full architectural case studies detailing the Problem, Solution, Technology Architecture, and National Impact.
Page 7: Contact Us	**NON-WORKING SUBMIT BUTTON** Visitors filling out the contact form could not submit it; clicking the button just refreshed the page and cleared their message.	**WORKING FORM WITH VALIDATION** Checks user inputs in real time, alerts if fields are missing, and shows a clear confirmation message upon submission.
Page 8: Verticals	**DUPLICATE TITLES & HEADINGS** All pages shared identical title tags; filler text ('Leadership Work') appeared in place of real technical capabilities.	**DEDICATED TECHNICAL DEPTH** Unique page titles, clear service descriptions, and detailed capability overviews for AI, Digital Identity, and Smart Cards.
Page 9: Custom 404	**RAW SERVER ERROR SCREEN** Mistyped links led to a plain server error screen with no navigation back to the website.	**BRANDED RECOVERY PAGE** Branded error page with helpful navigation, quick search, and an easy button to return home.

7. 46-Point Website Audit Status (Items 1 – 16)

Complete status tracking of the 46 specific problems identified during the initial audit. Issues marked **SOLVED** are fully repaired in code; items marked **NOT SOLVED** require external account access:

#	Problem Identified in Audit	Status	Resolution / Explanation
1	The contact form does not appear to work	SOLVED	Repaired with client-side validation, error handling, and instant confirmation.
2	Homepage banner buttons do nothing (Read More & Watch Video)	SOLVED	All 5 slide buttons linked to active service pages and official video showcase.
3	9 of 12 'read more' buttons on Services page do nothing	SOLVED	Replaced with 12 verified service cards with active destination links.
4	Website slides sideways on most laptops (400px overflow)	SOLVED	Removed 400px phone icon margin; fits 1366px screens smoothly with 0px wobble.
5	Website hard to use on mobile phones	SOLVED	Compact header, touch drawer menu (16px rows), clean whitish circuit banner.
6	Homepage banner can go blank for about 2 seconds	SOLVED	Synchronized all 5 carousel slides; restored Blockchain Technology slide.
7	Cloudflare security may block Google and AI tools	NOT SOLVED	Requires Cloudflare dashboard login to adjust Bot Fight Mode / WAF rules.
8	No sitemap to guide Google to our pages	SOLVED	Generated complete XML sitemap (sitemap.xml) covering all 9 pages.
9	All 9 pages have the same title and description	SOLVED	Wrote unique, keyword-rich meta titles and descriptions for every page.
10	8 of 9 pages have no main heading (H1) in code	SOLVED	Added one clear semantic H1 heading to every page for accessibility and SEO.
11	Company details not marked for Google/AI tools	SOLVED	Added Schema.org JSON-LD structured data (Organization & Contact point).
12	Links shared on social media have no preview image	SOLVED	Added OpenGraph and Twitter card preview images, titles, and descriptions.
13	Old page links show an error	SOLVED	Cleaned up URL mappings and cross-links to prevent 404 errors.
14	Pages do not clearly explain what company does	SOLVED	Expanded service descriptions, national projects showcase, and 3 case studies.
15	Homepage heavy at 2.48 MB, mostly images	SOLVED	Compressed and converted images to modern WebP format; page weight halved.
16	Duplicate and outdated code is loading	SOLVED	Consolidated styles and scripts into single, clean, minified production files.

8. 46-Point Website Audit Status (Items 17 – 31)

Continuing the 46-point checklist with detailed status and technical resolutions:

#	Problem Identified in Audit	Status	Resolution / Explanation
17	Company email easier to fake (no DMARC record)	NOT SOLVED	Requires domain DNS panel login to add TXT record _dmarc.spectrum.com.bd.
18	Standard website security settings off in Cloudflare	NOT SOLVED	Requires Cloudflare dashboard login to toggle HSTS and TLS 1.3 settings.
19	Images have no description (alt text)	SOLVED	Added descriptive alt text across all images on all pages.
20	Pale text is hard to read on all 9 pages	SOLVED	Replaced pale grey with high-contrast, comfortable #1e293b dark text.
21	Links have no name for screen readers	SOLVED	Added aria-label attributes to all icon links and interactive buttons.
22	Keyboard users cannot see which link is selected	SOLVED	Added visible high-contrast focus rings for keyboard tab navigation.
23	Footer background competes with the text	SOLVED	Softened background with dark overlay for clear reading contrast.
24	Banner moves automatically with no pause button	SOLVED	Added automatic pause-on-hover and touch swipe support.
25	Banner shows 5 dots for 4 slides (5th dot blank)	SOLVED	Synchronized 5 indicator dots to match the 5 real carousel slides.
26	Play icon on About picture does nothing	SOLVED	Connected to working video modal and official video channel link.
27	'Take a challenge!' looks like link but does nothing	SOLVED	Linked directly to the Contact / Consultation inquiry form.
28	Error page has no logo or menu	SOLVED	Built branded 404 recovery page with search bar, navigation, and home button.
29	Products and Contact pages have thin content	SOLVED	Expanded with product specifications, certification badges, and quote buttons.
30	Digital Identity URL contains space and '&'	SOLVED	Standardized clean address to dis.html with redirects from old path.
31	Project links ocd.org.bd / licence.gov.bd had errors	SOLVED	Verified and corrected destination government portal links.

9. 46-Point Website Audit Status (Items 32 – 46) & Summary

Final section of the 46-point checklist, followed by a clear summary of external requirements:

#	Problem Identified in Audit	Status	Resolution / Explanation
32	Links to ecourt / tcbsheba use old http addresses	SOLVED	Upgraded external links to secure HTTPS destinations where available.
33	Broken bullet point and long heading on AI page	SOLVED	Cleaned up formatting, shortened heading, and fixed bullet points.
34	Header phone number cannot be tapped to call	SOLVED	Added tel:+880... link so mobile visitors can tap to call immediately.
35	Some links are too small to tap on phones	SOLVED	Enlarged mobile touch targets to minimum 44px height.
36	Two large icon font files load for few icons	SOLVED	Replaced bloated icon font files with lightweight modern SVGs.
37	Contact form has seven required fields	SOLVED	Streamlined form; made secondary fields optional to improve conversions.
38	Phone box rejects '+880' and has no autofill	SOLVED	Configured tel input to accept international codes and browser autofill.
39	Phone number formatted differently across pages	SOLVED	Standardized phone format (+880-2-222281861) across all pages.
40	Some icons on Home and About look stretched	SOLVED	Fixed CSS object-fit to contain to preserve natural icon proportions.
41	Footer spelling: 'home & aboard'	SOLVED	Corrected spelling typo to 'home & abroad'.
42	Footer shows outdated year 2025	SOLVED	Updated copyright notice to the current year.
43	Case Study menu item has no page of its own	SOLVED	Created dedicated Case Studies page (case-study.html) with 3 client studies.
44	Banner labels 'Leadership Work' are vague	SOLVED	Replaced with verified ISO 9001 and ISO 27001 credentials.
45	Banner button 'Read More' gives no clear next step	SOLVED	Replaced with action-driven 'Explore Solutions' button.
46	Website is in English only (no Bangla)	NOT SOLVED	Architecture ready for dual-language; awaiting client-provided Bangla text.

Audit Resolution Summary:

- **42 of 46 Issues (91.3%) SOLVED:** All HTML, CSS, JavaScript, responsive layout, images, and content fixes are completely implemented and verified in plain-html/.
- **4 Issues NOT SOLVED (External Access Required):**
  1. **Items 7 & 18:** Require Spectrum Cloudflare dashboard login (to adjust Bot Fight Mode and toggle HSTS/TLS 1.3).
  2. **Item 17:** Requires domain DNS provider login (to publish email protection TXT DMARC record).
  3. **Item 46:** Multi-language structure is ready; awaiting client marketing copy for authentic Bangla text.